

# Chapter 91

# 3D Material Nodes

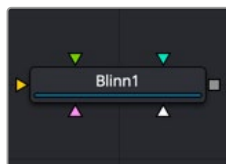
This chapter details the 3D Material nodes available when creating 3D composites in Fusion. The abbreviations next to each node name can be used in the Select Tool dialog when searching for tools and in scripting references.

For purposes of this document, node trees showing MediaIn nodes in DaVinci Resolve are interchangeable with Loader nodes in Fusion Studio, unless otherwise noted.

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# Blinn [3BI]



The Blinn node

## Blinn Node Introduction

The Blinn node is a basic illumination material that can be applied to geometry in the 3D scene. It describes how the object responds to light and provides multiple texture map inputs to allow fine control over the diffuse, specular, and bump map components of the material.

The standard basic material provided in the Materials tab of most geometry nodes is a simplified version of the Blinn node. The primary difference is that the Blinn node provides additional texture map inputs beyond just diffuse.

The Blinn node outputs a 3D Material that can be connected to the material inputs on any 3D geometry node.

The Blinn model in Fusion calculates the highlight as the dot product of the surface normal and the half angle vector between light source and viewer ( $\text{dot}(N, H)$ ). This may not always match the Blinn model illumination model used by other 3D applications.

## Inputs

There are five inputs on the Blinn node that accept 2D images or 3D materials. These inputs control the overall color and image used for the 3D object as well as the color and texture used in the specular highlight. Each of these inputs multiplies the pixels in the texture map by the equivalently named parameters in the node itself. This provides an effective method for scaling parts of the material.

**Diffuse Texture:** The orange Diffuse Texture input accepts a 2D image or a 3D material to be used as a main object texture map.

**Specular Color Material:** The green Specular Color material input accepts a 2D image or a 3D material to be used as the color texture map for specular highlight areas.

**Specular Intensity Materials:** The magenta Specular Intensity material input accepts a 2D image or a 3D material to be used to alter the intensity of specular highlights. When the input is a 2D image, the Alpha channel is used to create the map, while the color channels are discarded.

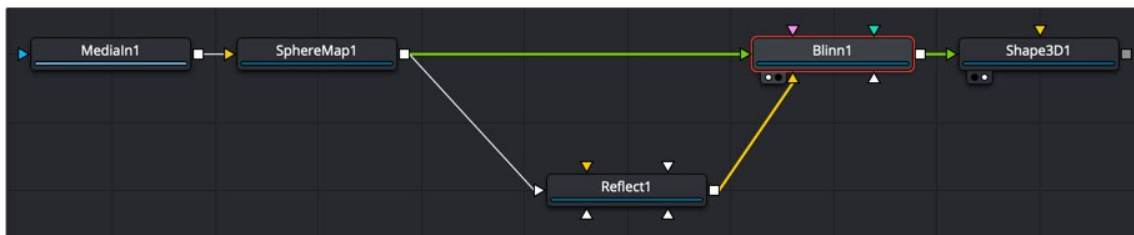
**Specular Exponent Material:** The teal Specular Exponent material input accepts a 2D image or a 3D material that is used as a falloff map for the material's specular highlights. When the input is a 2D image, the Alpha channel is used to create the map, while the color channels are discarded.

**Bump Map Material:** The white Bump Map material input accepts only a 3D material. Typically, you connect the texture into a Bump Map node, and then connect the Bump Map node to this input. This input uses the RGB information as texture-space normals.

When nodes have as many inputs as this one does, it is often difficult to make connections with any precision. Hold down the Option (macOS) or Alt (Windows) key while dragging the output from another node over the node tile, and keep holding Option or Alt when releasing the left mouse button. A small drop-down menu listing all the inputs provided by the node appears. Click on the desired input to complete the connection. Alternatively, you can drag the output from a node with the right mouse button to activate the same menu.

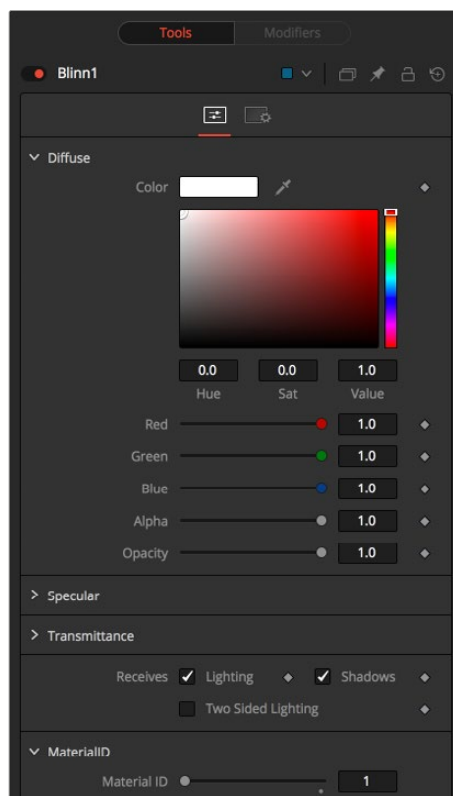
## Basic Node Setup

The output of a Blinn node output is connected to the material input on a 3D scene or 3D geometry node to which you want the shader applied. The Blinn inputs can use images as the diffuse color material (orange) and specular color material (green). This can lead to a smooth, shiny material.



A Blinn shader with diffuse and specular color materials connected

## Inspector



Blinn controls

## Controls Tab

The Controls tab is the primary tab for the Blinn node. It controls the color and shininess applied to the surface of the 3D geometry.

### Diffuse

Diffuse describes the base surface characteristics without any additional effects like reflections or specular highlights. Besides defining the base color of an object, the diffuse color also defines the transparency of the object. The Alpha in a diffuse texture map can be used to make portions of the surface transparent.

#### Diffuse Color

A material's Diffuse Color describes the base color presented by the material when it is lit indirectly or by ambient light. If a diffuse texture map is provided, then the color value provided here is multiplied by the color values in the texture.

#### Alpha

This slider sets the material's Alpha channel value. This affects diffuse and specular colors equally and affects the Alpha value of the material in the rendered output. If a diffuse texture map is provided, then the Alpha value set here is multiplied by the Alpha values in the texture map.

#### Opacity

Reducing the material's opacity decreases the color and Alpha values of the specular and diffuse colors equally, making the material transparent.

### Specular

The parameters in the Specular section describe the look of the specular highlight of the surface. These values are evaluated in a different way for each illumination model.

#### Specular Color

Specular Color determines the color of light that reflects from a shiny surface. The more specular a material is, the glossier it appears. Surfaces like plastics and glass tend to have white specular highlights, whereas metallic surfaces like gold have specular highlights that inherit their color from the material color. If a specular texture map is provided, then the value provided here is multiplied by the color values from the texture.

#### Specular Intensity

Specular Intensity controls how strong the specular highlight is. If the specular intensity texture is provided, then this value is multiplied by the Alpha value of the texture.

#### Specular Exponent

Specular Exponent controls the falloff of the specular highlight. The greater the value, the sharper the falloff, and the smoother and glossier the material appears. If the specular exponent texture is provided, then this value is multiplied by the Alpha value of the texture map.

## Transmittance

Transmittance controls the way light passes through a material. For example, a solid blue sphere casts a black shadow, but one made of translucent blue plastic would cast a much lower density blue shadow.

There is a separate Opacity option. Opacity determines how transparent the actual surface is when it is rendered. Fusion allows adjusting both opacity and transmittance separately. At first, this might be a bit counterintuitive to those who are unfamiliar with 3D software. It is possible to have a surface that is fully opaque but transmits 100% of the light arriving upon it, effectively making it a luminous/emissive surface.

## Attenuation

Attenuation determines how much color is passed through the object. For an object to have transmissive shadows, set the attenuation to (1, 1, 1), which means 100% of green, blue, and red light passes through the object. Setting this color to RGB (1, 0, 0) means that the material transmits 100% of the red arriving at the surface but none of the green or blue light. This can be used for “stained glass”-styled shadows.

## Alpha Detail

When the Alpha Detail slider is set to 0, the Alpha channel of the object is ignored and the entire object casts a shadow. If it is set to 1, the Alpha channel determines what portions of the object cast a shadow.

## Color Detail

The Color Detail slider modulates light passing through the surface by the diffuse color + texture colors. Use this to throw a shadow that contains color details of the texture applied to the object. Increasing the slider from 0 to 1 brings in more diffuse color + texture color into the shadow. Note that the Alpha and opacity of the object are ignored when transmitting color, allowing an object with a solid Alpha to still transmit its color to the shadow.

## Saturation

The Saturation slider controls the saturation of the color component transmitted to the shadow. Setting this to 0.0 results in monochrome shadows.

## Receives Lighting/Shadows

These checkboxes control whether the material is affected by lighting and shadows in the scene. If turned off, the object is always fully lit and/or unshadowed.

## Two-Sided Lighting

This effectively makes the surface two sided by adding a second set of normals facing the opposite direction on the backside of the surface. This is normally off to increase rendering speed, but it can be turned on for 2D surfaces or for objects that are not fully enclosed, to allow the reverse or interior surfaces to be visible as well.

Normally, in a 3D application, only the front face of a surface is visible and the back face is culled, so that if a camera were to revolve around a plane in a 3D application, when it reached the backside, the plane would become invisible. Making a plane two sided in a 3D application is equivalent to adding another plane on top of the first but rotated by 180 degrees so the normals are facing the opposite direction on the backside. Thus, when you revolve around the back, you see the second image plane, which has its normals facing the opposite way.

Fusion does exactly the same thing as 3D applications when you make a surface two sided. The confusion about what two-sided lighting does arises because Fusion does not cull back-facing polygons by default. If you revolve around a one-sided plane in Fusion, you still see it from the backside (but you are seeing the frontside duplicated through to the backside as if it were transparent). Making the plane two sided effectively adds a second set of normals to the backside of the plane.

**NOTE:** This can become rather confusing once you make the surface transparent, as the same rules still apply and produce a result that is counterintuitive. If you view from the frontside a transparent two-sided surface illuminated from the backside, it looks unlit.

### Material ID

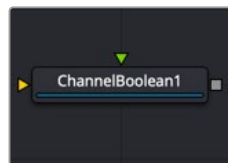
This slider sets the numeric identifier assigned to this material. This value is rendered into the MatID auxiliary channel if the corresponding option is enabled in the renderer.

### Common Controls

#### Settings Tab

The Settings tab in the Inspector is duplicated in other 3D nodes. These common controls are described in detail at the end of this chapter in “The Common Controls” section.

## Channel Boolean [3Boo]



The Channel Boolean node

### Channel Boolean Node Introduction

The Channel Boolean (not to be confused with the 2D Channel Booleans) can be used to remap and modify channels of 3D materials using mathematical operations. For example, if you want to use the red channel of a material to control a scalar input of an illumination model that uses the Alpha channel (e.g., Blinn. SpecularExponent), you can remap the channels here. Furthermore, it allows the use of geometry-specific information like texture space coordinates and normals.

### Inputs

There are two inputs on the Channel Boolean Node: one for the foreground material, and one for the background material. Both inputs accept either a 2D image or a 3D material like Blinn, Cook-Torrence, or Phong node.

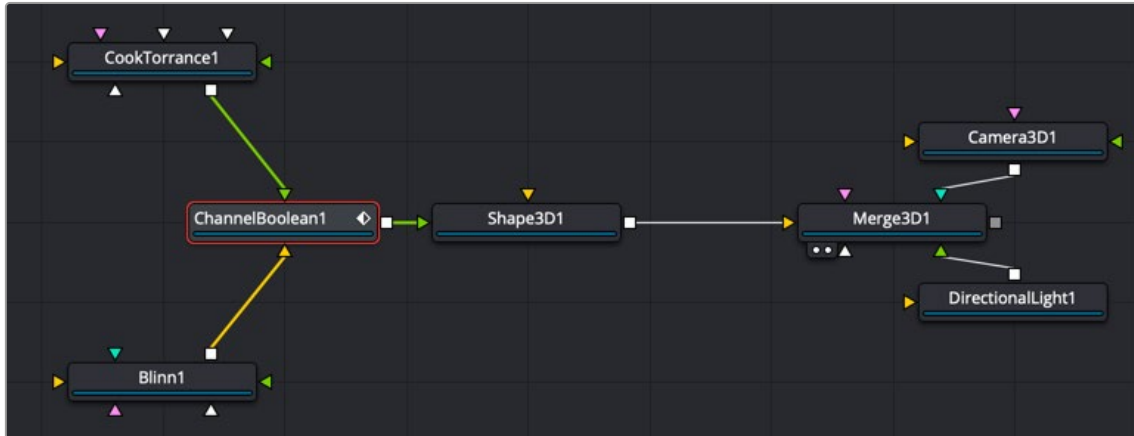
**BackgroundMaterial:** The orange background material input accepts a 2D image or a 3D material.

**ForegroundMaterial:** The green foreground input also accepts a 2D image or a 3D material.

## Basic Node Setup

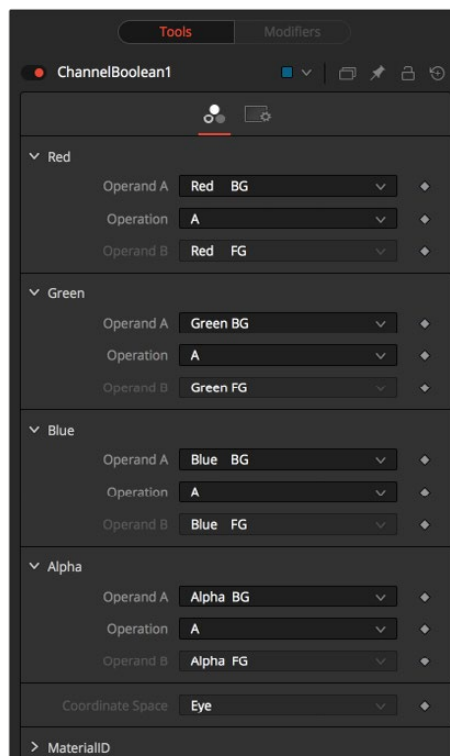
There are many uses for the material 3D Channel Boolean. Most often it is used to combine material looks or manipulate UV texture coordinates

In the below example, the Channel Boolean node combines the Cook Torrance and Blinn materials. It uses the math operands in the Channel Boolean to switch, invert, and mix the two inputs, creating a neon flickering effect.



A Channel Boolean used to combine and operate on Cook Torrance and Blinn nodes

## Inspector



Channel Boolean controls

## Controls Tab

The Controls tab includes a section for each RGBA channel. Within each channel are two input menus called Operand A and Operand B. The function performed on these two inputs is selected in the Operation menu.

### Operand A/B

The Operand menus, one for each output RGBA channel, allow you to set the desired input information for the corresponding channel.

|                                    |                                                                                                                                                                     |
|------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Red/Green/Blue/Alpha FG</b>     | Reads the color information of the foreground material.                                                                                                             |
| <b>Red/Green/Blue/Alpha BG</b>     | Reads the color information of the background material.                                                                                                             |
| <b>Black/White/Mid Gray</b>        | Sets the value of the channel to 0, 0.5, or 1.                                                                                                                      |
| <b>Hue/Lightness/Saturation FG</b> | Reads the color information of the foreground material, converts it into the HLS color space, and puts the selected information into the corresponding channel.     |
| <b>Hue/Lightness/Saturation BG</b> | Reads the color information of the background material, converts it into the HLS color space, and puts the selected information into the corresponding channel.     |
| <b>Luminance FG</b>                | Reads the color information of the foreground material and calculates the luminance value for the channel.                                                          |
| <b>Luminance BG</b>                | Reads the color information of the background material and calculates the luminance value for the channel.                                                          |
| <b>X/Y/Z Position FG</b>           | Sets the value of the channel to the position of the pixel in 3D space. The vector information is returned in eye space.                                            |
| <b>U/V/W Texture FG</b>            | Applies the texture space coordinates of the foreground material to the channels.                                                                                   |
| <b>U/V/W EnvCoords FG</b>          | Applies the environment texture space coordinates to the channels. Use it upstream of nodes modifying the environment texture coordinates like the Reflect 3D node. |
| <b>X/Y/Z Normal</b>                | Sets the value of the channel to the selected axis of the normal vector. The vector is returned in eye space.                                                       |



## Operation

Determines the Operation of how the operands are combined.

|                 |                                                                      |
|-----------------|----------------------------------------------------------------------|
| <b>A</b>        | Uses Operand A only for the output channel.                          |
| <b>B</b>        | Uses Operand B only for the output channel.                          |
| <b>1-A</b>      | Subtracts the value of Operand A from 1.                             |
| <b>1-B</b>      | Subtracts the value of Operand B from 1.                             |
| <b>A+B</b>      | Adds the value of Operand A and B.                                   |
| <b>A-B</b>      | Subtracts the value of Operand B from A.                             |
| <b>A*B</b>      | Multiplies the value of both Operands.                               |
| <b>A/B</b>      | Divides the value of Operand B from A.                               |
| <b>min(A,B)</b> | Compares the values of Operands A and B and returns the smaller one. |
| <b>max(A,B)</b> | Compares the values of Operands A and B and returns the bigger one.  |
| <b>avg(A,B)</b> | Returns the average value of both Operands.                          |

## Material ID

This slider sets the numeric identifier assigned to this material. This value is rendered into the MatID auxiliary channel if the corresponding option is enabled in the renderer.

## Common Controls

### Settings Tab

The Settings tab in the Inspector is duplicated in other 3D nodes. These common controls are described in detail at the end of this chapter in “The Common Controls” section.

# Cook Torrance [3CT]



The Cook Torrance node

## Cook Torrance Node Introduction

The Cook Torrance node is a basic illumination material that can be applied to geometry in the 3D scene. The diffuse calculation for this node is similar to that used in the basic material and the Blinn node, but the specular highlights are evaluated using an optimized Fresnel/Beckmann equation. This illumination model is primarily used for shading metal or other shiny and highly reflective surfaces.

The Cook Torrance node outputs a 3D Material that can be connected to the material inputs on any 3D geometry node.

## Inputs

There are six inputs on the Cook Torrance node that accept 2D images or 3D materials. These inputs control the overall color and image used for the 3D object as well as controlling the color and texture used in the specular highlight. Each of these inputs multiplies the pixels in the texture map by the equivalently named parameters in the node itself. This provides an effective method for scaling parts of the material.

**Diffuse Color Material:** The orange Diffuse Color material input accepts a 2D image or a 3D material to be used as overall color and texture of the object.

**Specular Color Material:** The green Specular Color material input accepts a 2D image or a 3D material to be used as the color and texture of the specular highlight.

**Specular Intensity Material:** The magenta Specular Intensity material input accepts a 2D image or a 3D material to alter the intensity of the specular highlight. When the input is a 2D image, the Alpha channel is used to create the map, while the color channels are discarded.

**Specular Roughness Material:** The white Specular Roughness material input accepts a 2D image or a 3D material to be used as a map for modifying the roughness of the specular highlight. The Alpha of the texture map is multiplied by the value of the roughness control.

**Specular Refractive Index Material:** The white Specular Refractive Index material input accepts a 2D image or a 3D material, using the RGB channels as the refraction texture.

**Bump Map Material:** The white Bump Map material input accepts only a 3D material. Typically, you connect the texture into a Bump Map node, and then connect the Bump Map node to this input. This input uses the RGB information as texture-space normals.

Each of these inputs multiplies the pixels in the texture map by the equivalently named parameters in the node itself. This provides an effective method for scaling parts of the material.

When nodes have as many inputs as this one does, it is often difficult to make connections with any precision. Hold down the Option (macOS) or Alt (Windows) key while dragging the output from another node over the node tile, and keep holding Option or Alt when releasing the left mouse button. A small drop-down menu listing all the inputs provided by the node appears. Click on the desired input to complete the connection.

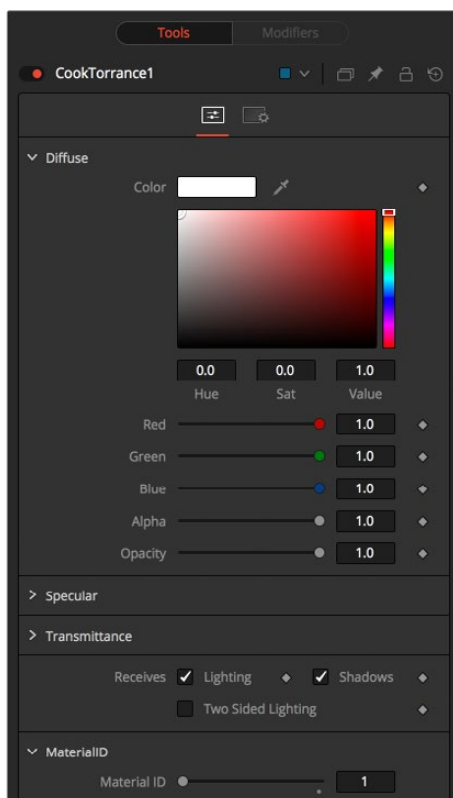
## Basic Node Setup

The output of a Cook Torrance node output is connected to the material input on a 3D scene or 3D geometry node to which you want the shader applied. The Cook Torrance inputs can use images as the diffuse color material (yellow) and specular color material (green). This can result in a smooth, shiny material.



A Cook Torrance shader with diffuse and specular color materials connected

## Inspector



Cook Torrance controls

### Controls Tab

The Controls tab contains parameters for adjusting the main color, highlight, and lighting properties of the Cook Torrance shader node.

### Diffuse

Diffuse describes the base surface characteristics without any additional effects like reflections or specular highlights. Besides defining the base color of an object, the diffuse color also defines the transparency of the object. The Alpha in a diffuse texture map can be used to make portions of the surface transparent.

#### Diffuse Color

A material's Diffuse Color describes the base color presented by the material when it is lit indirectly or by ambient light. If a diffuse texture map is provided, then the color value provided here is multiplied by the color values in the texture.

#### Alpha

This slider sets the material's Alpha channel value. This affects diffuse and specular colors equally, and affects the Alpha value of the material in the rendered output. If a diffuse texture map is provided, then the Alpha value set here is multiplied by the Alpha values in the texture map.

#### Opacity

Reducing the material's Opacity decreases the color and Alpha values of the specular and diffuse colors equally, making the material transparent.

## Specular

The parameters in the Specular section describe the look of the specular highlight of the surface. These values are evaluated in a different way for each illumination model.

### Specular Color

Specular Color determines the color of light that reflects from a shiny surface. The more specular a material is, the glossier it appears. Surfaces like plastics and glass tend to have white specular highlights, whereas metallic surfaces like gold have specular highlights that inherit their color from the material color. If a specular texture map is provided, then the value provided here is multiplied by the color values from the texture.

### Specular Intensity

Specular Intensity controls how strong the specular highlight is. If the specular intensity texture is provided, then this value is multiplied by the Alpha value of the texture.

### Roughness

The Roughness of the specular highlight describes diffusion of the specular highlight over the surface. The greater the value, the wider the falloff, and the more brushed and metallic the surface appears. If the roughness texture map is provided, then this value is multiplied by the Alpha value from the texture.

### Do Fresnel

Selecting this checkbox adds Fresnel calculations to the materials illumination model. This provides more realistic-looking metal surfaces by taking into account the refractiveness of the material.

### Refractive Index

This slider appears when the Do Fresnel checkbox is selected. The Refractive Index applies only to the calculations for the highlight; it does not perform actual refraction of light through transparent surfaces. If the refractive index texture map is provided, then this value is multiplied by the Alpha value of the input.

## Transmittance

Transmittance controls the way light passes through a material. For example, a solid blue sphere casts a black shadow, but one made of translucent blue plastic would cast a much lower density blue shadow.

There is a separate Opacity option. Opacity determines how transparent the actual surface is when it is rendered. Fusion allows adjusting both opacity and transmittance separately. At first, this might be a bit counterintuitive to those who are unfamiliar with 3D software. It is possible to have a surface that is fully opaque but transmits 100% of the light arriving upon it, effectively making it a luminous/emissive surface.

### Attenuation

Attenuation determines how much color is passed through the object. For an object to have transmissive shadows, set the attenuation to (1, 1, 1), which means 100% of green, blue, and red light passes through the object. Setting this color to RGB (1, 0, 0) means that the material transmits 100% of the red arriving at the surface but none of the green or blue light. This can be used to create “stained glass”-styled shadows.

### Alpha Detail

When the Alpha Detail slider is set to 0, the Alpha channel of the object is ignored and the entire object casts a shadow. If it is set to 1, the Alpha channel determines what portions of the object cast a shadow.

### Color Detail

The Color Detail slider modulates light passing through the surface by the diffuse color + texture colors. Use this to throw a shadow that contains color details of the texture applied to the object. Increasing the slider from 0 to 1 brings in more diffuse color + texture color into the shadow. Note that the Alpha and opacity of the object are ignored when transmitting color, allowing an object with a solid Alpha to still transmit its color to the shadow.

### Saturation

The Saturation slider controls the saturation of the color component transmitted to the shadow. Setting this to 0.0 results in monochrome shadows.

### Receives Lighting/Shadows

These checkboxes control whether the material is affected by lighting and shadows in the scene. If turned off, the object is always fully lit and/or unshadowed.

### Two-Sided Lighting

This effectively makes the surface two sided by adding a second set of normals facing the opposite direction on the backside of the surface. This is normally off to increase rendering speed, but it can be turned on for 2D surfaces or for objects that are not fully enclosed, to allow the reverse or interior surfaces to be visible as well.

Normally, in a 3D application, only the front face of a surface is visible and the back face is culled, so that if a camera were to revolve around a plane in a 3D application, when it reached the backside, the plane would become invisible. Making a plane two sided in a 3D application is equivalent to adding another plane on top of the first but rotated by 180 degrees so the normals are facing the opposite direction on the backside. Thus, when you revolve around the back, you see the second image plane, which has its normals facing the opposite way.

**NOTE:** This can become rather confusing once you make the surface transparent, as the same rules still apply and produce a result that is counterintuitive. If you view from the frontside a transparent two-sided surface illuminated from the backside, it looks unlit.

### Material ID

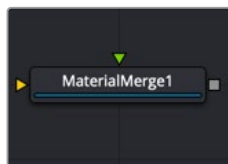
This slider sets the numeric identifier assigned to this material. This value is rendered into the MatID auxiliary channel if the corresponding option is enabled in the renderer.

## Common Controls

### Settings Tab

The Settings tab in the Inspector is duplicated in other 3D nodes. These common controls are described in detail at the end of this chapter in “The Common Controls” section.

# Material Merge 3D [3MM]



The Material Merge node

## Material Merge Node Introduction

The Material Merge node can be used to combine two separate materials together. This node can be used to composite Material nodes, combining multiple illumination materials (Blinn, Cook Torrance) with texture nodes (Bump Map, Reflection) to create complex shader networks.

The node also provides a mechanism for assigning a new material identifier to the combined material.

## Inputs

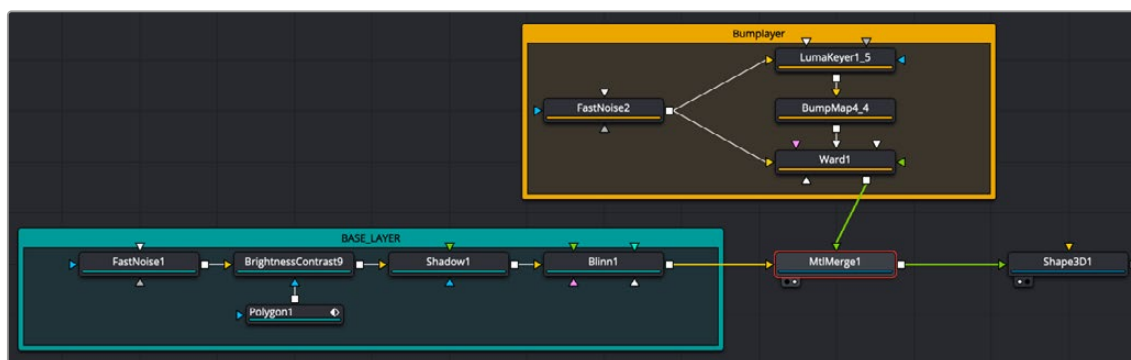
The Material Merge node includes two inputs for the two materials you want to combine.

**Background Material:** The orange Background material input accepts a 2D image or a 3D material to be used as the background material.

**Foreground Material:** The green Foreground material input accepts a 2D image or a 3D material to be used as the foreground material. A 2D image is treated as a diffuse texture map in the basic shading model.

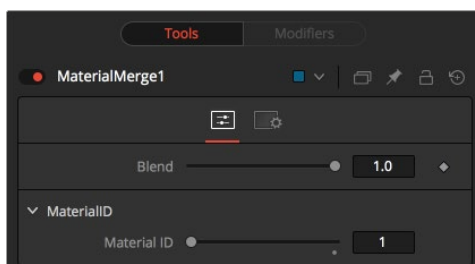
## Basic Node Setup

The output of a Material Merge node is connected to the material input on a 3D scene or 3D geometry node. The Material Merge node below is taking in a background base layer from the Blinn shader and combining it with a more textured bump map layer.



A Material Merge node combining a Blinn-based shader (teal underlay) and a Ward-based shader (orange underlay)

## Inspector



Material Merge controls

### Controls Tab

The Controls tab includes a single slider for blending the two materials together.

### Blend

The Blend behavior of the Material Merge is similar to the Dissolve (DX) node for images. The two materials/textures are mixed using the value of the slider to determine the percentage each input contributes. While the background and foreground inputs can be a 2D image instead of a material, the output of this node is always a material.

Unlike the 2D Dissolve node, both foreground and background inputs are required.

### Material ID

This slider sets the numeric identifier assigned to the resulting material. This value is rendered into the MatID auxiliary channel if the corresponding option is enabled in the renderer.

### Common Controls

#### Settings Tab

The Settings tab in the Inspector is duplicated in other 3D nodes. These common controls are described in detail at the end of this chapter in “The Common Controls” section.

## Phong [3Ph]



The Phong node

### Phong Node Introduction

The Phong node is a basic illumination material that can be applied to geometry in the 3D scene. It describes how the object responds to light and provides multiple texture map inputs to allow fine control over the diffuse, specular, and bump map components of the material.

While producing a highlight similar to that produced by the Blinn model, it is more commonly used for shiny/polished plastic surfaces.

## Inputs

There are five inputs on the Phong node that accept 2D images or 3D materials. These inputs control the overall color and image used for the 3D object as well as controlling the color and texture used in the specular highlight. Each of these inputs multiplies the pixels in the texture map by the equivalently named parameters in the node itself. This provides an effective method for scaling parts of the material.

**Diffuse Material:** The orange Diffuse material input accepts a 2D image or a 3D material to be used as a main color and texture of the object.

**Specular Color Material:** The green Specular Color material input accepts a 2D image or a 3D material to be used as a highlight color and texture of the object.

**Specular Intensity Material:** The magenta Specular Intensity material input accepts a 2D image or a 3D material to be used as an intensity map for the material's highlights. When the input is a 2D image, the Alpha channel is used to create the map, while the color channels are discarded.

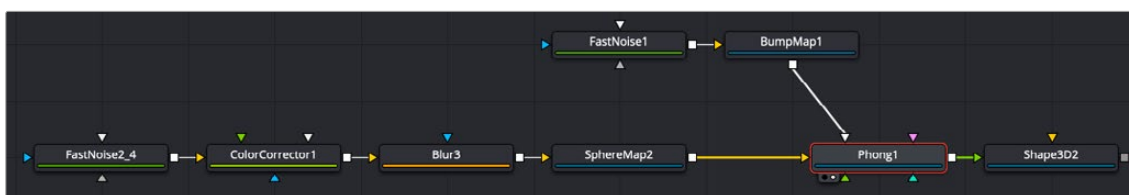
**Specular Exponent Material:** The teal Specular Exponent material input accepts a 2D image or a 3D material to be used as a falloff map for the material's specular highlights. When the input is a 2D image, the Alpha channel is used to create the map, while the color channels are discarded.

**Bump Map Material:** The white Bump Map texture input accepts only a 3D material. Typically, you connect the texture into a Bump Map node, and then connect the Bump Map node to this input. This input uses the RGB information as texture-space normals.

When nodes have as many inputs as this one does, it is often difficult to make connections with any precision. Hold down the Option or Alt key while dragging the output from another node over the node tile, and keep holding Option or Alt when releasing the left mouse button. A small drop-down menu listing all the inputs provided by the node appears. Click on the desired input to complete the connection.

## Basic Node Setup

The output of a Phong node is connected to the material input on a 3D scene or 3D geometry node. The Phong node below is taking in a base Color Diffuse input from the Fast Noise node and a bump input map texture also generated from a Fast Noise node.



A Phong node with a diffuse color and Bump Map input



## Inspector



Phong controls

### Controls Tab

The Controls tab contains parameters for adjusting the main color, highlight, and lighting properties of the Phong shader node.

### Diffuse

Diffuse describes the base surface characteristics without any additional effects like reflections or specular highlights. Besides defining the base color of an object, the diffuse color also defines the transparency of the object.

The Alpha in a diffuse texture map can be used to make portions of the surface transparent.

#### Diffuse Color

A material's Diffuse Color describes the base color presented by the material when it is lit indirectly or by ambient light. If a diffuse texture map is provided, then the color value provided here is multiplied by the color values in the texture.

#### Alpha

This slider sets the material's Alpha channel value. This affects diffuse and specular colors equally and affects the Alpha value of the material in the rendered output. If a diffuse texture map is provided, then the Alpha value set here is multiplied by the Alpha values in the texture map.

#### Opacity

Reducing the material's Opacity decreases the color and Alpha values of the specular and diffuse colors equally, making the material transparent.

## Specular

The parameters in the Specular section describe the look of the specular highlight of the surface. These values are evaluated in a different way for each illumination model.

### Specular Color

Specular Color determines the color of light that reflects from a shiny surface. The more specular a material is, the glossier it appears. Surfaces like plastics and glass tend to have white specular highlights, whereas metallic surfaces like gold have specular highlights that inherit their color from the material color. If a specular texture map is provided, then the value provided here is multiplied by the color values from the texture.

### Specular Intensity

Specular Intensity controls how strong the specular highlight is. If the specular intensity texture is provided, then this value is multiplied by the Alpha value of the texture.

### Specular Exponent

Specular Exponent controls the falloff of the specular highlight. The greater the value, the sharper the falloff, and the smoother and glossier the material appears. If the specular exponent texture is provided, then this value is multiplied by the Alpha value of the texture map.

## Transmittance

Transmittance controls the way light passes through a material. For example, a solid blue sphere casts a black shadow, but one made of translucent blue plastic would cast a much lower density blue shadow.

There is a separate Opacity option. Opacity determines how transparent the actual surface is when it is rendered. Fusion allows adjusting both opacity and transmittance separately. At first, this might be a bit counterintuitive to those who are unfamiliar with 3D software. It is possible to have a surface that is fully opaque but transmits 100% of the light arriving upon it, effectively making it a luminous/emissive surface.

### Attenuation

Attenuation determines how much color is passed through the object. For an object to have transmissive shadows, set the attenuation to (1, 1, 1), which means 100% of green, blue, and red light passes through the object. Setting this color to RGB (1, 0, 0) means that the material transmits 100% of the red arriving at the surface but none of the green or blue light. This can be used to create “stained glass”-styled shadows.

### Alpha Detail

When the Alpha Detail slider is set to 0, the Alpha channel of the object is ignored and the entire object casts a shadow. If it is set to 1, the Alpha channel determines what portions of the object cast a shadow.

### Color Detail

The Color Detail slider modulates light passing through the surface by the diffuse color + texture colors. Use this to throw a shadow that contains color details of the texture applied to the object. Increasing the slider from 0 to 1 brings in more diffuse color + texture color into the shadow. Note that the Alpha and opacity of the object are ignored when transmitting color, allowing an object with a solid Alpha to still transmit its color to the shadow.

### Saturation

The Saturation slider controls the saturation of the color component transmitted to the shadow. Setting this to 0.0 results in monochrome shadows.

### Receives Lighting/Shadows

These checkboxes control whether the material is affected by lighting and shadows in the scene. If turned off, the object is always fully lit and/or unshadowed.

### Two-Sided Lighting

This effectively makes the surface two sided by adding a second set of normals facing the opposite direction on the backside of the surface. This is normally off to increase rendering speed, but it can be turned on for 2D surfaces or for objects that are not fully enclosed, to allow the reverse or interior surfaces to be visible as well.

Normally, in a 3D application, only the front face of a surface is visible and the back face is culled, so that if a camera were to revolve around a plane in a 3D application, when it reached the backside, the plane would become invisible. Making a plane two sided in a 3D application is equivalent to adding another plane on top of the first but rotated by 180 degrees so the normals are facing the opposite direction on the backside. Thus, when you revolve around the back, you see the second image plane, which has its normals facing the opposite way.

Fusion does exactly the same thing as 3D applications when you make a surface two sided. The confusion about what two-sided lighting does arises because Fusion does not cull back-facing polygons by default. If you revolve around a one-sided plane in Fusion, you still see it from the backside (but you are seeing the frontside duplicated through to the backside as if it were transparent). Making the plane two sided effectively adds a second set of normals to the backside of the plane.

**NOTE:** This can become rather confusing once you make the surface transparent, as the same rules still apply and produce a result that is counterintuitive. If you view from the frontside a transparent two-sided surface illuminated from the backside, it looks unlit.

### Material ID

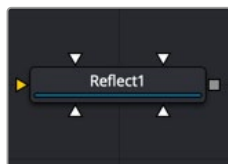
This slider sets the numeric identifier assigned to this material. This value is rendered into the MatID auxiliary channel if the corresponding option is enabled in the renderer.

## Common Controls

### Settings Tab

The Settings tab in the Inspector is duplicated in other 3D nodes. These common controls are described in detail at the end of this chapter in “The Common Controls” section.

# Reflect [3Rr]



The Reflect node

## Reflect Node Introduction

The Reflect node is used to add environment map reflections and refractions to materials.

Control is offered over the face on and glancing strength, falloff, per channel refraction indexes, and tinting. Several texture map inputs can modify the behavior of each parameter.

Environment mapping is an approximation that assumes an object's environment is infinitely distant from the object. It's best to picture this as a cube or sphere with the object at the center. Specifically, this infinite distance assumption means that objects cannot interact with themselves (e.g., the reflections on the handle of a teapot do not show the body of the teapot but rather the infinite environment map). It also means that if you use the same cube map on multiple objects in the scene, those objects do not inter-reflect each other (e.g., two neighboring objects would not reflect each other). If you want objects to reflect each other, you need to render a cube map for each object.

For more information, see *Chapter 85, "3D Compositing Basics,"* in the DaVinci Resolve Reference Manual, or Chapter 25 in the Fusion Reference Manual.

## Inputs

There are five inputs on the Reflect node that accept 2D images or 3D materials. These inputs control the overall color and image used for the 3D object as well as controlling the color and texture used in the reflective highlights.

**Background Material:** The orange Background material input accepts a 2D image or a 3D material. If a 2D image is provided, the node treats it as a diffuse texture map applied to a basic material.

**Reflection Color Material:** The white Reflection Color material input accepts a 2D image or a 3D material. The RGB channels are used as the reflection texture, and the Alpha is ignored.

**Reflection Intensity Material:** The white Reflection Intensity material input accepts a 2D image or a 3D material. The Alpha channel of the texture is multiplied by the intensity of the reflection.

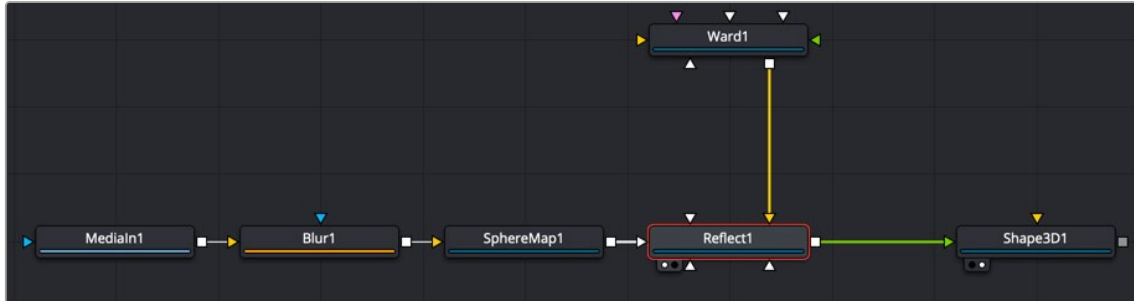
**Refraction Tint Material:** The white Refraction Tint material input accepts a 2D image or a 3D material. The RGB channels are used as the refraction texture.

**Bump Map Texture:** The white Bump Map texture input accepts only a 3D material. Typically, you connect the texture into a Bump Map node, and then connect the Bump Map node to this input. This input uses the RGB information as texture-space normals.

When nodes have as many inputs and some using the same color as this one does, it is often difficult to make connections with any precision. Hold down the Option or Alt key while dragging the output from another node over the node tile, and keep holding Option or Alt when releasing the left mouse button. A small drop-down menu listing all the inputs provided by the node appears. Click on the desired input to complete the connection.

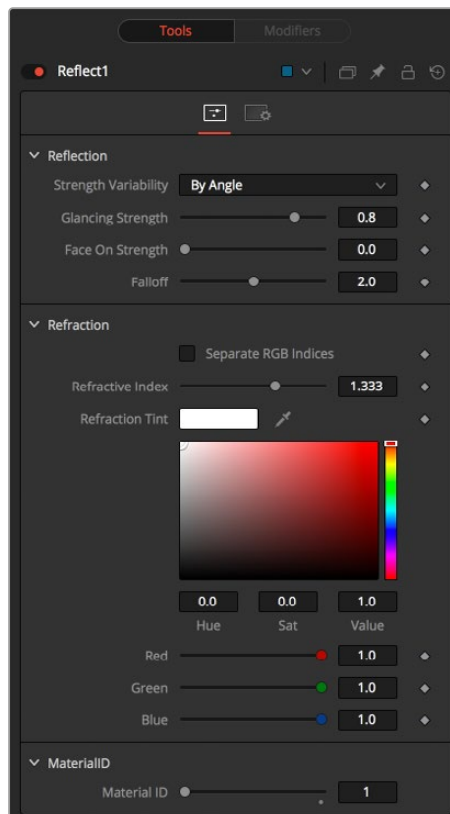
## Basic Node Setup

The Reflection node can be the main shader for an object as it is in the example below, or it can be used to feed the diffuse material input of a Ward, Blinn, Phong, or other material node. Usually, a Sphere Map node is used as the source of the Reflect node's reflection color input.



A Reflect node used to create a highly reflective surface of a Shape 3D node

## Inspector



Reflect controls

## Controls Tab

The Controls tab contains parameters for adjusting the reflective strength based on the orientation of the object, as well as the tint color of the Reflect shader node.

## Reflection

### Reflection Strength Variability

This multi-button control can be set to Constant or By Angle for varying the reflection intensity, corresponding to the relative surface orientation to the viewer. The following three controls are visible only when this control is set to By Angle.

### Glancing Strength

[By Angle] Glancing Strength controls the intensity of the reflection for those areas of the geometry where the reflection faces away from the camera.

### Face On Strength

[By Angle] Face On Strength controls the intensity of the reflection for those parts of the geometry that reflect directly back to the camera.

### Falloff

[By Angle] Falloff controls the sharpness of the transition between the Glancing and Face On Strength regions. It can be considered similar to applying gamma correction to a gradient between the Face On and Glancing values.

### Constant Strength

[Constant Angle] This control is visible only when the reflection strength variability is set to Constant. In this case, the intensity of the reflection is constant despite the incidence angle of the reflection.

## Refraction

If the incoming background material has a lower opacity than 1, then it is possible to use an environment map as refraction texture, and it is possible to simulate refraction effects in transparent objects.

### Separate RGB Refraction Indices

When this checkbox is enabled, the Refraction Index slider is hidden, and three sliders for adjusting the refraction index of the Red, Green, and Blue channels appear in its place. This allows simulation of the spectral refraction effects commonly seen in thick imperfect glass, for example.

### Refraction Index

This slider controls how strongly the environment map is deformed when viewed through a surface. The overall deformation is based on the incidence angle. Since this is an approximation and not a simulation, the results are not intended to model real refractions accurately.

### Refraction Tint

The refraction texture is multiplied by the tint color for simulating color-filtered refractions. It can be used to simulate the type of coloring found in tinted glass, as seen in many brands of beer bottles, for example.

## Common Controls

### Settings Tab

The Settings tab in the Inspector is duplicated in other 3D nodes. These common controls are described in detail at the end of this chapter in “The Common Controls” section.

# Stereo Mix [3SMM]



The Stereo Mix node

## Stereo Mix Node Overview

This node is used to swap the left and right material inputs. It is often used to output to the left and right eye of the 3D Render.

### Inputs

This node has two inputs that are both required for this node to work. Both inputs accept either a 2D image or a 3D material.

**LeftMaterial:** The orange left material input accepts a 2D image or a 3D material to be used as the material for the left eye rendering. If a 2D image is used, it is converted to a diffuse texture map using the basic material type.

**RightMaterial:** The green right material input accepts a 2D image or a 3D material to be used as the material for the right eye rendering. If a 2D image is used, it is converted to a diffuse texture map using the basic material type.

While the inputs can be either 2D images or 3D materials, the output is always a material.

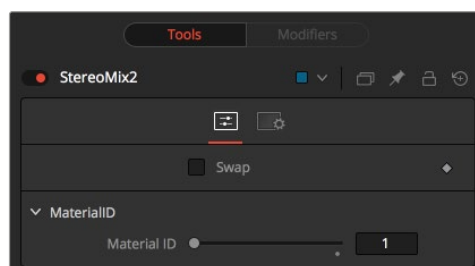
## Basic Node Setup

The Stereo Mix node can be used with either stereo images or materials. The example below shows two images combined in the Stereo Mix node causing the output to be a stereo anaglyph material.



A Stereo Mix node used to combine left and right images into a single stereo material

## Inspector



Stereo Mix controls

## Controls Tab

The Controls tab contains a single switch that swaps the left and right material inputs.

### Swap

This option swaps both inputs of the node.

### Material ID

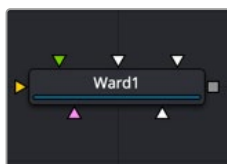
This slider sets the numeric identifier assigned to this material. This value is rendered into the MatID auxiliary channel if the corresponding option is enabled in the renderer.

## Common Controls

### Settings Tab

The Settings tab in the Inspector is duplicated in other 3D nodes. These common controls are described in detail at the end of this chapter in “The Common Controls” section.

## Ward [3Wd]



The Ward node

## Ward Node Introduction

The Ward node is a basic illumination material that can be applied to geometry in the 3D scene. It describes how the object responds to light and provides multiple texture map inputs to allow fine control over the diffuse, specular, and bump map components of the material.

Specifically, the Ward node is ideal for simulating brushed metal surfaces, as the highlight can be elongated along the U or V directions of the mapping coordinates. This is known as an anisotropic highlight.

The Ward node outputs a 3D Material that can be connected to the material inputs on any 3D geometry node.

## Inputs

There are six inputs on the Ward node that accept 2D images or 3D materials. These inputs control the overall color and image used for the 3D object as well as controlling the color and texture used in the specular highlight. Each of these inputs multiplies the pixels in the texture map by the equivalently named parameters in the node itself. This provides an effective method for scaling parts of the material.

**Diffuse Material:** The orange Diffuse material input accepts a 2D image or a 3D material to be used as a main color and texture of the object.



**Specular Color Material:** The green Specular Color material input accepts a 2D image or a 3D material to be used as a highlight color and texture of the object.

**Specular Intensity Material:** The magenta Specular Intensity material input accepts a 2D image or a 3D material to be used as an intensity map for the material's highlights. When the input is a 2D image, the Alpha channel is used to create the map, while the color channels are discarded.

**Spread U Material:** The white Spread U material input accepts a 2D image or a 3D material. The value of the Spread U option in the node's controls is multiplied against the pixel values in the material's Alpha channel.

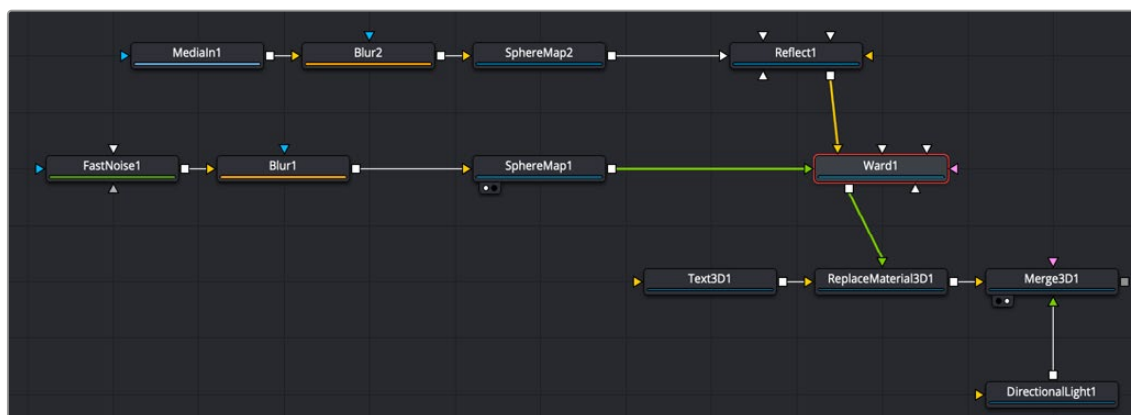
**Spread V Material:** The white Spread V material input accepts a 2D image or a 3D material. The value of the Spread V option in the node's controls is multiplied against the pixel values in the material's Alpha channel.

**Bump Map Material:** The white Bump Map material input accepts only a 3D material. Typically, you connect the texture into a Bump Map node, and then connect the Bump Map node to this input. This input uses the RGB information as texture-space normals.

When nodes have as many inputs and some using the same color as this one does, it is often difficult to make connections with any precision. Hold down the Option or Alt key while dragging the output from another node over the node tile, and keep holding Option or Alt when releasing the left mouse button. A small drop-down menu listing all the inputs provided by the node appears. Click on the desired input to complete the connection.

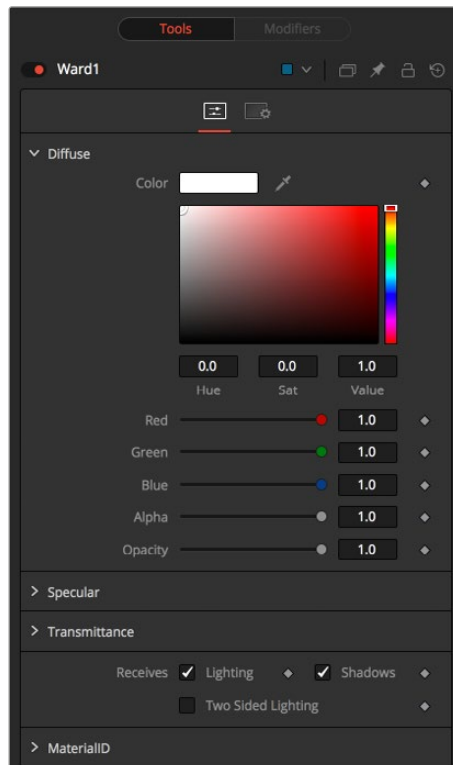
## Basic Node Setup

The Ward node is used to make a shiny glass surface and replace the 3D text material in the example below. A diffuse color material comes from Reflect node, and the specular color is altered by a gradient color Fast Noise node.



A Ward node used with a diffuse connection and specular color connection

## Inspector



Ward controls

### Controls Tab

The Controls tab contains parameters for adjusting the main color, highlight, and lighting properties of the Ward shader node.

#### Diffuse

Diffuse describes the base surface characteristics without any additional effects like reflections or specular highlights. Besides defining the base color of an object, the diffuse color also defines the transparency of the object. The Alpha in a diffuse texture map can be used to make portions of the surface transparent.

#### Diffuse Color

A material's Diffuse Color describes the base color presented by the material when it is lit indirectly or by ambient light. If a diffuse texture map is provided, then the color value provided here is multiplied by the color values in the texture.

#### Alpha

This slider sets the material's Alpha channel value. This affects diffuse and specular colors equally and affects the Alpha value of the material in the rendered output. If a diffuse texture map is provided, then the Alpha value set here is multiplied by the Alpha values in the texture map.

#### Opacity

Reducing the material's Opacity decreases the color and Alpha values of the specular and diffuse colors equally, making the material transparent.

## Specular

The parameters in the Specular section describe the look of the specular highlight of the surface. These values are evaluated in a different way for each illumination model.

### Specular Color

Specular Color determines the color of light that reflects from a shiny surface. The more specular a material is, the glossier it appears. Surfaces like plastics and glass tend to have white specular highlights, whereas metallic surfaces like gold have specular highlights that inherit their color from the material color. If a specular texture map is provided, then the value provided here is multiplied by the color values from the texture.

### Specular Intensity

Specular Intensity controls how strong the specular highlight is. If the specular intensity texture is provided, then this value is multiplied by the Alpha value of the texture.

### Spread U

Spread U controls the falloff of the specular highlight along the U-axis in the UV map of the object. The smaller the value, the sharper the falloff, and the smoother and glossier the material appears in this direction. If the Spread U texture is provided, then this value is multiplied by the Alpha value of the texture.

### Spread V

Spread V controls the falloff of the specular highlight along the V-axis in the UV map of the object. The smaller the value, the sharper the falloff, and the smoother and glossier the material appear in this direction. If the Spread V texture is provided, then this value is multiplied by the Alpha value of the texture.

## Transmittance

Transmittance controls the way light passes through a material. For example, a solid blue sphere casts a black shadow, but one made of translucent blue plastic would cast a much lower density blue shadow.

There is a separate Opacity option. Opacity determines how transparent the actual surface is when it is rendered. Fusion allows adjusting both opacity and transmittance separately. At first, this might be a bit counterintuitive to those who are unfamiliar with 3D software. It is possible to have a surface that is fully opaque but transmits 100% of the light arriving upon it, effectively making it a luminous/emissive surface.

### Attenuation

Attenuation determines how much color is passed through the object. For an object to have transmissive shadows, set the attenuation to (1, 1, 1), which means 100% of green, blue, and red light passes through the object. Setting this color to RGB (1, 0, 0) means that the material transmits 100% of the red arriving at the surface but none of the green or blue light. This can be used to create “stained glass”-styled shadows.

### Alpha Detail

When the Alpha Detail slider is set to 0, the Alpha channel of the object is ignored, and the entire object casts a shadow. If it is set to 1, the Alpha channel determines what portions of the object cast a shadow.

## Color Detail

The Color Detail slider modulates light passing through the surface by the diffuse color + texture colors. Use this to throw a shadow that contains color details of the texture applied to the object. Increasing the slider from 0 to 1 brings in more diffuse color + texture color into the shadow. Note that the Alpha and opacity of the object are ignored when transmitting color, allowing an object with a solid Alpha to still transmit its color to the shadow.

## Saturation

The Saturation slider controls the saturation of the color component transmitted to the shadow. Setting this to 0.0 results in monochrome shadows.

## Receives Lighting/Shadows

These checkboxes control whether the material is affected by lighting and shadows in the scene. If turned off, the object is always fully lit and/or unshadowed.

## Two-Sided Lighting

This effectively makes the surface two sided by adding a second set of normals facing the opposite direction on the backside of the surface. This is normally off to increase rendering speed, but it can be turned on for 2D surfaces or for objects that are not fully enclosed, to allow the reverse or interior surfaces to be visible as well.

Normally, in a 3D application, only the front face of a surface is visible and the back face is culled, so that if a camera were to revolve around a plane in a 3D application, when it reached the backside, the plane would become invisible. Making a plane two sided in a 3D application is equivalent to adding another plane on top of the first but rotated by 180 degrees so the normals are facing the opposite direction on the backside. Thus, when you revolve around the back, you see the second image plane, which has its normals facing the opposite way.

Fusion does exactly the same thing as 3D applications when you make a surface two sided. The confusion about what two-sided lighting does arises because Fusion does not cull back-facing polygons by default. If you revolve around a one-sided plane in Fusion you still see it from the backside (but you are seeing the frontside duplicated through to the backside as if it were transparent). Making the plane two sided effectively adds a second set of normals to the backside of the plane.

**NOTE:** This can become rather confusing once you make the surface transparent, as the same rules still apply and produce a result that is counterintuitive. If you view from the frontside a transparent two-sided surface illuminated from the backside, it looks unlit.

## Material ID

This slider sets the numeric identifier assigned to this material. This value is rendered into the MatID auxiliary channel if the corresponding option is enabled in the renderer.

## Common Controls

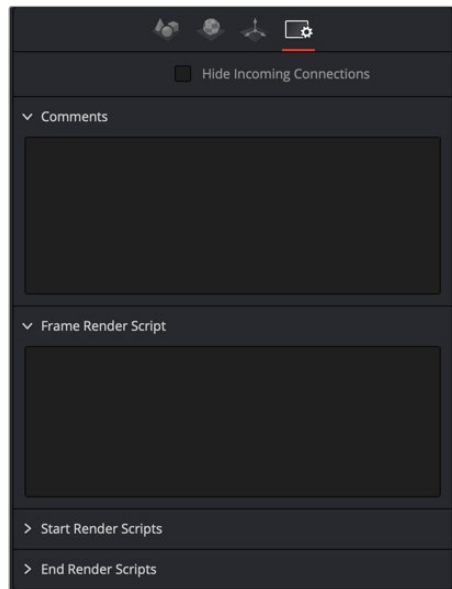
### Settings Tab

The Settings tab in the Inspector is duplicated in other 3D nodes. These common controls are described in detail in the following “The Common Controls” section.

# The Common Controls

Nodes that handle 3D geometry share a number of identical controls in the Inspector. This section describes controls that are common among 3D Material nodes.

## Settings Tab



Common Settings 3D controls

Common Settings tab can be found on most tools in Fusion. The following controls are specific settings for 3D nodes.

### Hide Incoming Connections

Enabling this checkbox can hide connection lines from incoming nodes, making a node tree appear cleaner and easier to read. When enabled, fields for each input on a node are displayed. Dragging a connected node from the node tree into the field hides that incoming connection line as long as the node is not selected in the node tree. When the node is selected in the node tree, the line reappears.

### Comment Tab

The Comment tab contains a single text control that is used to add comments and notes to the tool. When a note is added to a tool, a small red dot icon appears next to the setting's tab icon, and a text bubble appears on the node. To see the note in the Node Editor, hold the mouse pointer over the node for a moment. The contents of the Comments tab can be animated over time, if required.

### Scripting Tab

The Scripting tab is present on every tool in Fusion. It contains several edit boxes used to add scripts that process when the tool is rendering. For more details on the contents of this tab, please consult the scripting documentation.